First idea: With the file we were given, we think the words “dwarf” and “giant” represent 1’s and 0’s. We thought about modifying the Binary Decoder program we made to work with this. An easier solution we found was to search and replace the words with 0’s and 1’s. After running this through the decoder, we got:

UGK hvievi lm 138.47.98.222
fhvi: vxdzrjp
kzhh: xxxgbzoku

Lu xlfihv, gsv zxgfzo fhvimznv zmw kzhhdliw nfhg yv wvxibkgvw gl ol:

After translating through an Atbash cipher decoder, we got this:
[Atbash Cipher - Reverse Mirror Alphabet - Online Decoder/Translator]
FTP server on 138.47.98.222
user: ecwaiqk
pass: ccctyalpf

Of course, the actual username and password must be decrypted to lo:

We tried finding the decrypted username and password by brute-forcing using a Vigenere cipher:
[Vigenere Cipher Solver - Online Decoder, Encoder, Translator]

Possible Users:	inthere

WPDTEZG	inthere
LVSJEIX	therein
LVSJEQR	thereat
LVSJADE	thering
LVWHADE	thating
LVSJIDH	therand
LVSJEQS	thereas
LVSJEDR	therent
WPDTEYR	inthest
LVSIPIX	thestin
WPDTEB	inthepo
WPDTEK	inthego

CYDTEZ	cetheri
WPDTEX	intheto
WPDTEN	inthedo
MCDHB	satthei

Possible Pass:	ontherear

UPJMUJHXM	intherest
JVYCUISHS	therestin
UPJMUJHWY	inthereth
UPJMUISIB	inthesthe
UPJMUJSIB	intherthe
JVYCUIDCZ	theresing
JVYCUISBA	therestof
JVYCUASHS	thereatin
UPJMUJHXN	intheress
JVYCUNSBA	therentof
YLYBFWUW	eresterth
OPJMUHEL	onthether
YLYBFWTW	erestesth
OPJMUJHP	ontherear
YLYBFMTW	erestosth
OPJMUJHK	ontherefr
OPJMUJHA	otherepr
POJMUJH	notherear
OPJMUJHW	ontheretr
OJVPHWGB	otherefor
OPJMUJSI	ontherthr
POJMUIH	nothesear

If you use Polaris, from the challenge text callsign, as the key for the vigenere cipher, and input the password ccctyalpf you get northstar.

Assuming the username is the callsign too, we were able to log into the FTP server.

After that, we listed the files and found a directory called “files”. In that were two more directories called “10” and “7”. In each directory is a list of files that we are going to run through our FTP covert channel program, assuming the names of the directories are the number of bits being used.

After modifying the FTP Program to accept the IP, username, and password, the following was decoded from the permission strings:
7:
grains of sand on all the beaches of Earth.
10:
Even the darkest nebula hides new suns

After coming back to it a few minutes later, other messages were discovered

7:
cosmos to know itself.
ang and will end in silence.
rsation between mass and spacetime.
is on the side of the outcast.
ity; it is a profound source of spirituality.
very star it has ever touched.
a sun to someone.
, and leads us from this world to another.
; sunlight is painting.
erse, the universe is in us.
ible is waiting to be known.
furnace burning for billions of years.
d you touch a child.
he carbon in our apple pies were made in the interiors of collapsing stars.
atters would be to limit the human spirit.
eeping the dark with patient regularity.
g. Curiosity has its own reason for existing.
really. It's an imperative.
just us, seems like an awful waste of space.
ff. We are a way for the universe to know itself.
God divided by zero.

10:

Gravity is not a force but a conve
The universe began with a b
Science is not only compatible with spiritual
At the heart of every star is a nuclear
Moonlight is sculpture
Not only are we in the univ
Light carries the memory of e
Every star may be
Black holes are where

The messages can be strung together as:

Gravity is not a force but a conversation between mass and spacetime.
Science is not only compatible with spirituality; it is a profound source of spirituality.
The universe began with a bang and will end in silence.
Light carries the memory of every star it has ever touched
Every star may be a sun to someone.
Not only are we in the universe, the universe is in us.
Black holes are where God divided by zero.
Moonlight is sculpture; sunlight is painting.
Our sun is one of a hundred billion stars in our galaxy. Our galaxy is one of billions of galaxies populating the universe.
Two possibilities exist: either we are alone in the universe or we are not. Both are equally terrifying.

REALIZED WE ONLY NEED ONE

Astronomy compels the soul to look upward, and leads us from this world to another.

- 7:05 PM 10/12/26